

Audio Management System—a Digital Signal Processor (DSP) whose responsibility it is to send the right frequencies of sound to the right speakers—splitting the sound into bass, mids, and high frequencies. A seemingly simple task, but as Sunil explains, “Lower frequencies tend to move speaker cones a lot, so bass must go to the bigger, heavier speakers built to handle this movement.” And what happens if we were to try playing bass through the other, lighter speakers? “They’d blow up...” Clearly not a fitting end to crores worth of equipment.

The DSP also adjusts sound volumes in such a way that speakers meant for the front of the hall deliver a softer sound than the speakers meant for the back. This is called *Amplitude Shading*, and this is why your head doesn’t explode when you’re in the first row. To control all these settings, the SOUND.COM guys use *Audio Core*—XTA’s own software designed to work with the DSPs they manufacture.

To analyse the sound in the hall, they use SIA’s *SmaartLive*, which measures the difference between the sound that is supposed to come out of the speakers (the electric signal sent to it from the DSP) and the sound that actually does. *SmaartLive* and *Audio Core* interact with each other, so when *SmaartLive* detects any problems with the sound, *Audio Core* can be used to compensate.

An Equal Music

Now that the speakers were positioned, and everyone was convinced they weren’t going to topple over and kill innocent people, it was time to configure them for the Ultimate Music Experience we keep hankering after.

The first hindrance to the said experience is the fact that sound travels slowly, so if you happen to be towards the end of the hall,



“Even with better technology, nothing sounds as good as the older vacuum-tube guitar amps. We actually use one that was built in 1959!”

Sunil Karanjikar
Systems/Mix Engineer
SOUND.COM



These speakers at the front of the stage ensure that the front rows aren’t left out

you’re first going to hear the sound from the second set of speakers we mentioned, and then, albeit softer, you’re going to hear the sound that came from the speakers at the front of the hall. Up front, too, you’ll face a similar situation—you’ll first hear the sound from the speakers, and then from the instruments themselves—all extremely unpleasant. To get around this, *Audio Core* delays the sound going to all the speakers so that all sounds hit your ears at the same time.

We took a trip to the back to see this in action, and sure enough, we first heard a drum beat from the speaker closest to us, quickly followed by another from the speakers close to the stage—a decidedly unpleasant echo, echo, echo... Put simply, if you say the infamous “Mic Check, one, two, three,” the audience at the back would hear, “Mic(Mic) check(check) one(one), two(two), thr(th)ee(ree)”. And it sounds worse than it looks, too.

There was a microphone near the speakers at the stage, and another next to the second set near the end of the hall, and they recorded the time it took for sound to travel from the stage to the back. This information was then fed to *Audio Core*, and soon enough, the delay

